

ALBERTUS, IL, USA

WMO: 722082

Lat: 42.246N

Lon: 89.582W

Elev: 859

StdP: 14.25

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04876

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-1.0	-14.3	2.6	-5.7	-9.0	3.5	0.4	26.5	25.3	24.7	26.6	5.8	270	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	89.1	74.9	86.1	73.0	83.6	72.2	78.3	85.5	76.1	83.2	74.1	81.0	11.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	142.5	83.7	73.5	128.5	80.5	72.0	122.4	78.9	42.4	85.2	40.1	83.2	38.3	81.2	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	20.8	18.7	-13.7	92.8	7.0	3.9	-18.8	95.6	-22.8	97.9	-26.7	100.1	-31.8	103.0
			-14.3	80.9	6.5	2.8	-19.0	83.0	-22.9	84.6	-26.5	86.2	-31.3	88.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	21.1	23.1	36.3	48.4	59.1	68.6	71.5	69.4	63.4	50.7	37.9	26.0
	DBStd	20.19	12.19	11.78	11.30	8.99	9.04	5.81	6.09	5.43	7.64	9.15	9.87	10.86
	HDD50	3483	896	755	449	134	19	0	0	0	3	104	378	745
	HDD65	6812	1361	1174	892	501	229	30	12	19	119	453	813	1209
	CDD50	2784	0	1	23	87	302	559	665	601	405	125	15	1
	CDD65	638	0	0	1	4	47	139	212	155	70	10	0	0
	CDH74	5834	0	0	12	64	478	1221	2016	1302	638	103	0	0
	CDH80	1555	0	0	0	6	118	314	636	314	152	15	0	0
	WSAvg	8.2	9.4	9.2	9.3	10.4	9.0	7.6	6.0	5.4	6.5	8.2	9.3	9.0
	PrecAvg	36.9	1.6	1.5	2.3	3.6	4.2	5.2	3.8	4.2	3.6	2.6	2.6	1.7
Precipitation	PrecMax	45.4	3.3	3.7	4.8	6.1	8.9	11.7	7.6	9.4	8.0	5.3	5.4	4.3
	PrecMin	27.0	0.5	0.2	0.6	1.6	0.6	1.5	0.7	0.9	0.9	0.4	0.0	0.5
	PrecStd	4.6	0.7	1.0	1.5	1.5	2.2	2.7	1.5	2.1	1.7	1.6	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.6	58.7	76.5	80.2	88.8	90.5	95.2	91.0	88.9	81.8	67.8	57.8
		MCWB	47.9	51.3	62.5	62.4	69.7	74.6	80.5	74.9	73.9	68.2	57.6	55.2
	2%	DB	43.8	47.8	64.1	75.0	82.6	86.6	90.3	86.2	84.2	77.0	61.8	48.9
		MCWB	40.7	43.7	56.4	59.9	66.6	72.3	78.2	74.2	71.2	63.9	54.6	46.0
	5%	DB	39.2	41.5	59.3	69.7	79.5	83.8	86.1	83.2	81.0	71.8	57.3	43.9
		MCWB	36.5	37.7	52.0	55.6	65.7	70.9	74.5	72.8	68.4	61.9	51.8	41.4
	10%	DB	36.6	37.5	53.2	64.3	75.4	81.0	83.5	81.1	77.5	66.3	53.2	39.4
		MCWB	34.8	35.0	47.0	54.1	63.9	68.9	73.4	71.1	66.4	59.3	48.0	37.0
	0.4%	WB	48.9	52.2	63.9	66.2	73.6	78.6	82.4	79.4	76.3	71.7	60.3	55.2
		MCDB	50.3	56.2	74.3	75.7	84.1	86.9	91.5	88.0	84.1	79.4	64.6	58.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.2	43.7	58.4	61.5	70.7	75.6	79.6	76.7	73.9	67.8	55.9	46.0
		MCDB	43.8	47.6	64.5	70.8	79.7	83.3	86.8	83.0	80.7	73.8	60.6	47.7
	5%	WB	37.3	38.5	53.5	58.8	68.2	73.5	77.3	74.5	71.4	63.6	52.4	41.5
		MCDB	39.1	40.9	58.6	67.5	76.4	80.8	84.2	80.8	77.8	68.5	56.5	43.5
	10%	WB	34.5	35.4	47.3	55.5	65.6	71.4	74.8	72.6	69.1	59.7	48.1	37.5
		MCDB	36.2	37.8	53.5	63.6	73.3	77.9	81.7	78.8	74.9	66.1	53.1	39.5
Mean Daily Temperature Range	5% DB	MDBR	13.9	14.8	16.7	20.9	20.5	19.9	19.8	19.8	21.7	19.5	15.9	13.1
		MCDBR	15.7	18.4	24.4	29.0	24.6	22.5	21.0	21.3	24.0	25.7	21.0	16.6
		MCWBR	13.7	14.4	16.9	16.9	13.0	12.2	11.0	11.9	12.7	15.0	15.5	14.3
	5% WB	MCDBR	15.3	18.1	22.0	24.9	21.0	19.8	18.5	18.1	20.3	20.9	19.7	16.1
		MCWBR	13.5	14.7	16.6	16.5	12.8	12.6	10.9	11.8	12.3	15.6	15.9	14.5
Clear-Sky Solar Irradiance	taub		0.282	0.300	0.332	0.380	0.407	0.422	0.439	0.415	0.387	0.339	0.298	0.281
	taud		2.397	2.365	2.392	2.272	2.254	2.254	2.200	2.266	2.324	2.458	2.507	2.434
	Ebn at Noon		275	287	290	281	276	271	264	267	267	270	268	266
	Edh at Noon		25	30	34	41	43	43	45	41	36	28	23	22
All-Sky Solar Radiation	RadAvg		550	862	1183	1472	1721	1944	1933	1705	1413	923	601	442
	RadStd		66	103	115	122	115	147	113	107	115	112	51	41

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page